

Foglio 3 - Soluzioni,

ESERCIZIO 1

Calcolare l'integrale di superficie

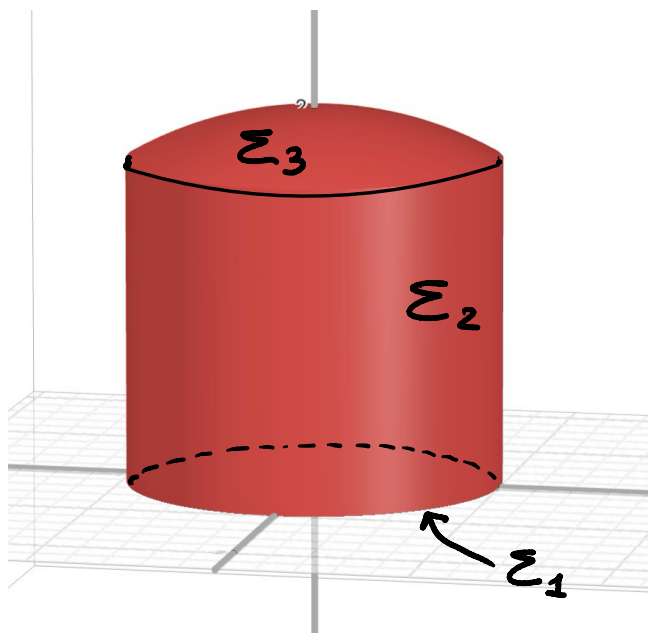
$$\int_{\partial T} y \, d\sigma$$

dove $T = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 \leq 4, x^2 + y^2 \leq 1, z \geq 0\}$

Soluzione

$$T = \{(x, y, z) \in \mathbb{R}^3 \mid \underbrace{x^2 + y^2 + z^2 \leq 4}_{\text{sfera centrata in } (0,0,0) \text{ di raggio } 2}, \underbrace{x^2 + y^2 \leq 1}_{\text{cilindro centrato su asse } z, \text{ raggio } 1}, z \geq 0\}$$

$$\partial T = \Sigma_1 \cup \Sigma_2 \cup \Sigma_3$$



Paramettrizziamo Σ_i , $i=1,2,3$

$$\bullet) \Sigma_1 = \varphi_1(D_1) \text{ con } \varphi_1: D_1 = [0,1] \times [0,2\pi] \longrightarrow \mathbb{R}^3$$

$$(p, \theta) \longmapsto (p \cos \theta, p \sin \theta, 0)$$

$$\bullet) \Sigma_2 = \varphi_2(D_2) \text{ con } \varphi_2: D_2 = [0,h] \times [0,2\pi] \longrightarrow \mathbb{R}^3$$

$$(z, \theta) \longmapsto (\cos \theta, \sin \theta, z)$$

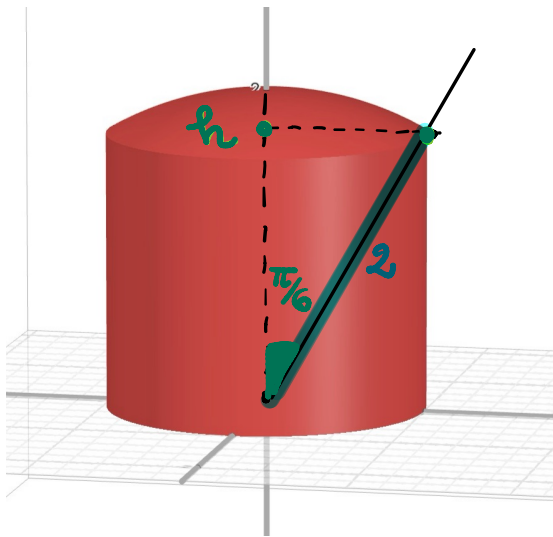
Per trovare h , interseco superficie sfera e superficie cilindro (+ condizione $h \geq 0$):

$$\begin{cases} x^2 + y^2 + h^2 = 4 \\ x^2 + y^2 = 1 \\ h \geq 0 \end{cases} \implies h = \sqrt{3} \rightsquigarrow D_2 = [0, \sqrt{3}] \times [0, 2\pi]$$

angolo limite: $2 \cos \theta = h = \sqrt{3} \implies \theta = \pi/6$

$$\bullet) \Sigma_3 = \varphi_3(D_3) \text{ con } \varphi_3: D_3 = [0, \pi/6] \times [0, 2\pi] \longrightarrow \mathbb{R}^3$$

$$(\theta, \varphi) \longmapsto (2 \sin \theta \cos \varphi, 2 \sin \theta \sin \varphi, 2 \cos \theta)$$



$$I := \int_{\partial T} y \, d\tau = \sum_{i=1}^3 \int_{\Sigma_i} y \, d\tau_i$$

Calcoliamo i 3 addendi nel RHS usando le nostre parametrizzazioni.

$$\bullet) \int_{\Sigma_1} y d\sigma_1 = (*)$$

$$d\sigma_1 = \left\| \frac{\partial \varphi_1}{\partial p} \times \frac{\partial \varphi_1}{\partial \theta} \right\| dp d\theta = \left\| \frac{\partial \varphi_1}{\partial p} \right\| \left\| \frac{\partial \varphi_1}{\partial \theta} \right\| dp d\theta = 1 \cdot p dp d\theta = p dp d\theta$$

$$\left\langle \frac{\partial \varphi_1}{\partial p}, \frac{\partial \varphi_1}{\partial \theta} \right\rangle = 0 \quad (\text{check!})$$

$$(*) = \int_0^1 p^2 dp \underbrace{\int_0^{2\pi} \sin \theta d\theta}_{=0} = 0$$

$$\bullet) \int_{\Sigma_2} y d\sigma_2 = (*)$$

$$d\sigma_2 = \left\| \frac{\partial \varphi_2}{\partial z} \times \frac{\partial \varphi_2}{\partial \theta} \right\| dz d\theta = \left\| \frac{\partial \varphi_2}{\partial z} \right\| \left\| \frac{\partial \varphi_2}{\partial \theta} \right\| dz d\theta = 1 \cdot 1 dz d\theta = dz d\theta$$

$$\left\langle \frac{\partial \varphi_2}{\partial z}, \frac{\partial \varphi_2}{\partial \theta} \right\rangle = 0 \quad (\text{check!})$$

$$(*) = \int_0^{\sqrt{3}} dz \int_0^{2\pi} \sin \theta d\theta = 0$$

- $\int_{\Sigma_3} y d\sigma_3 = (*)$

$$d\sigma_3 = \left\| \frac{\partial \varphi_3}{\partial \theta} \times \frac{\partial \varphi_3}{\partial \varphi} \right\| d\theta d\varphi \stackrel{\uparrow}{=} \left\| \frac{\partial \varphi_3}{\partial \theta} \right\| \left\| \frac{\partial \varphi_3}{\partial \varphi} \right\| d\theta d\varphi \stackrel{(\text{check!})}{=} 4 \sin \theta d\theta d\varphi$$

$$\left\langle \frac{\partial \varphi_3}{\partial \theta}, \frac{\partial \varphi_3}{\partial \varphi} \right\rangle = 0 \quad (\text{check!})$$

$$(*) = \int_0^{\pi/6} 8 \sin \theta d\theta \int_0^{2\pi} \sin \varphi d\varphi = 0$$

$$\Rightarrow I = 0$$

Trick $(x, y, z) \in \partial T \iff (x, -y, z) \in \partial T$
 $f(x, y, z) = -y = -f(x, y, z)$ $\Rightarrow I = 0$ (no conti :))

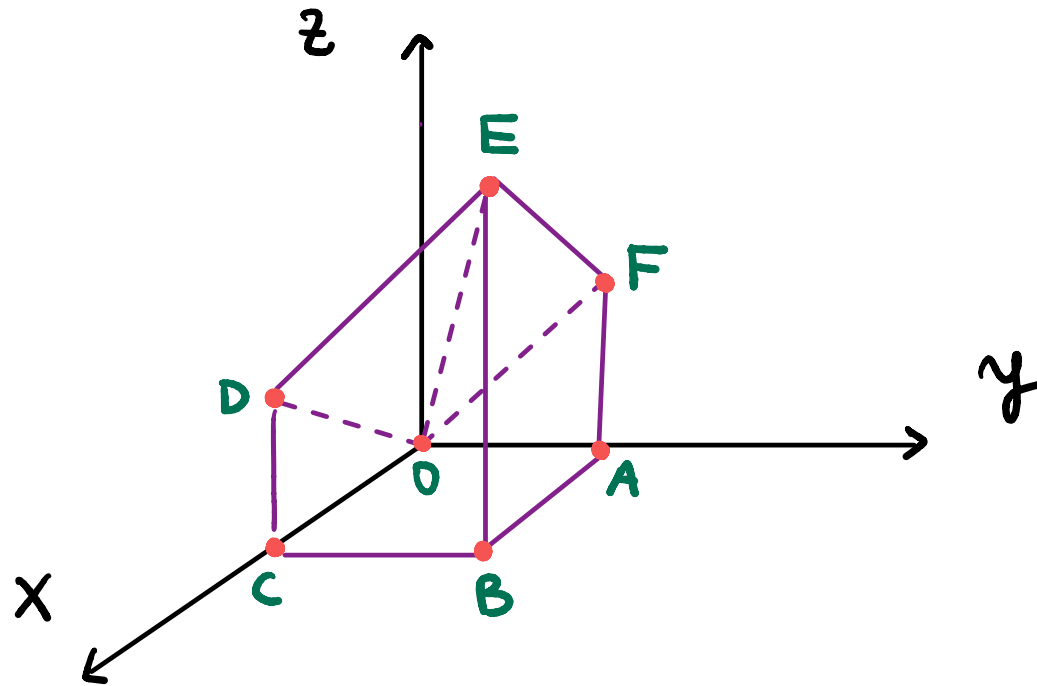
ESERCIZIO 2 (Diano del corso, es 49)

Sia Ω il poliedro di vertici $(0,0,0)$, $(0,1,0)$, $(1,1,0)$, $(1,0,0)$, $(1,0,1)$, $(1,1,2)$, $(0,1,1)$.

Calcolare

$$I := \iiint_{\Omega} \frac{1}{(z+1)^3} dx dy dz$$

Soluzione



E, D, F, O sono complanari. Il piano π che li contiene è dato, parametricamente

da
$$\begin{cases} x = t \\ y = s \\ z = t + s \end{cases} ; t, s \in \mathbb{R}, \quad \xrightarrow{\text{eq. cartesiana}} \quad \pi: z = x + y$$

Dunque: $\Omega = \{(x, y, z) \in \mathbb{R}^3 \mid (x, y) \in [0, 1]^2, 0 \leq z \leq x+y\}$

Ω è (x, y) -normale. Quindi

$$I = \int_0^1 \int_0^1 \left(\int_0^{x+y} \frac{dz}{(z+1)^3} \right) dx dy$$

$$= \int_0^1 \int_0^1 \left[-\frac{1}{2(z+1)^2} \right]_0^{x+y} dx dy$$

$$= \int_0^1 \int_0^1 \left(\frac{1}{2} - \frac{1}{2(x+y+1)^2} \right) dx dy$$

$$= \frac{1}{2} - \frac{1}{2} \int_0^1 \int_0^1 \frac{1}{(x+y+1)^2} dx dy$$

$$= \frac{1}{2} - \frac{1}{2} \int_0^1 \left[-\frac{1}{(x+y+1)} \right]_0^1 dy$$

$$= \frac{1}{2} - \frac{1}{2} \int_0^1 \left(\frac{1}{y+1} - \frac{1}{y+2} \right) dy$$

$$= \frac{1}{2} - \frac{1}{2} \left[\log \left| \frac{y+1}{y+2} \right| \right]_0^1 =$$

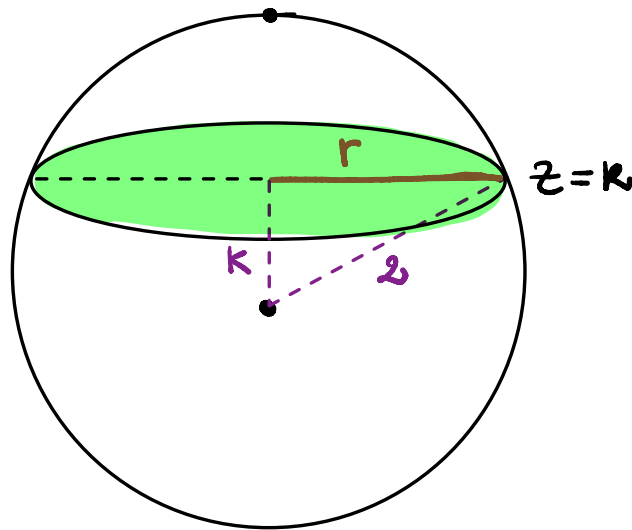
$$= \frac{1}{2} - \frac{1}{2} \log \frac{2}{3} + \frac{1}{2} \log \frac{1}{2} = \frac{1}{2} \left(1 - \log \left(\frac{4}{3} \right) \right)$$

ESERCIZIO 3

Calcolare

$$I := \iiint_E \frac{1}{\sqrt{x^2+y^2}} dx dy dz$$

dove $E = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 \leq 4, z \geq 1\}$



Tagliando la calotta sferica con il piano $z=k \in [1,2]$, otteniamo i cerchi $x^2 + y^2 \leq r^2$ dove $r = \sqrt{4-z^2}$, cosicché $r \in [0, \sqrt{3}]$

Procedendo per sezioni, si ha:

$$\iiint_E \frac{1}{\sqrt{x^2+y^2}} dx dy dz = \int_1^2 \left(\iint_{\{x^2+y^2 \leq 4-z^2\}} \frac{1}{\sqrt{x^2+y^2}} dx dy \right) dz$$

$$\text{POLARI su } \{x^2+y^2 \leq 4-z^2\} = \int_1^2 \left(\int_0^{2\pi} \int_0^{\sqrt{4-z^2}} \rho d\theta \right) dz$$

$$= 2\pi \int_1^2 \sqrt{4-z^2} dz$$

$$= 4\pi \int_1^2 \sqrt{1-(z/2)^2} dz$$

$$\begin{aligned} \sinh u &= (z/2) \\ \cosh u du &= 1/2 dz \\ u &= \operatorname{arcsinh}(z/2) \end{aligned}$$

$$= 4\pi \int_{\pi/6}^{\pi/2} 2 \cosh^2 u du$$

$$= 8\pi \int_{\pi/6}^{\pi/2} \cosh^2 u du$$

$$\int \cosh^2 u du = \frac{u + \cosh u \sinh u}{2} + C = 4\pi \left[\frac{\pi}{2} - \frac{\pi}{6} - \frac{\sqrt{3}}{4} \right] = \frac{4}{3}\pi - \sqrt{3}\pi$$

[Provare per fili!]

ESERCIZIO 4

Calcolare in 3 diversi modi l'area di

$$\Pi = \{(x, y, z) \in \mathbb{R}^3 \mid (\sqrt{x^2 + y^2} - 2)^2 + (z - 1)^2 = 1\}$$

Soluzione Si tratta di un toro. Per vederlo, basta porre $r = \sqrt{x^2 + y^2}$.

L'eq diventa $(r - 2)^2 + (z - 1)^2 = 1$, cioè un cerchio nel piano rz di raggio 1 e centro $(r, z) = (2, 1)$. Ruotando tale cerchio attorno all'asse z , troviamo Π

Metodo 1: Pappo-Guldino

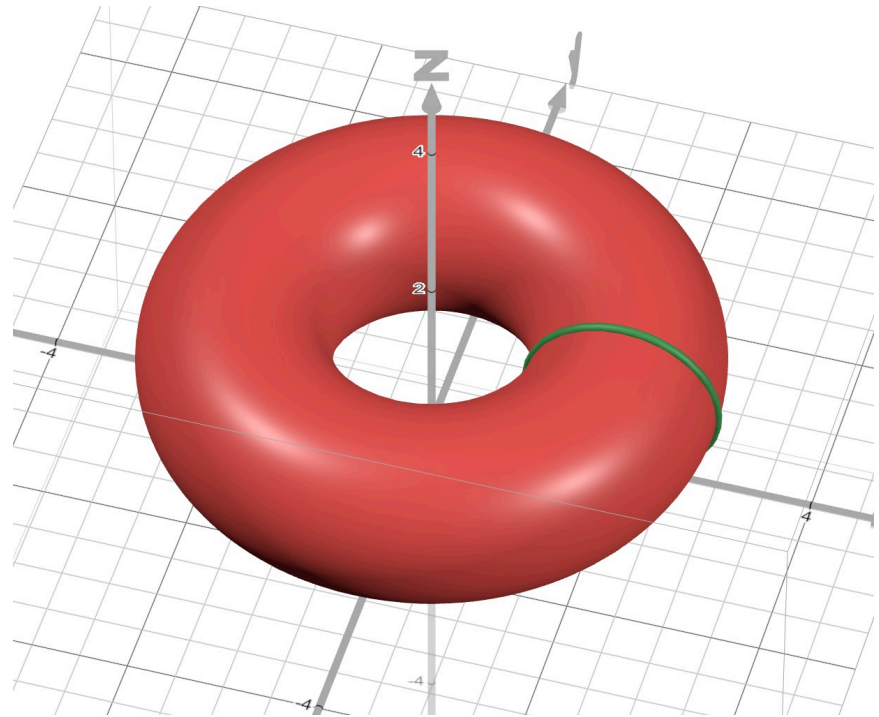
Π può essere def. come rotazione di 2π attorno all'asse z della curva

$$\gamma(\theta) = (2 + \cos \theta, 0, 1 + \sin \theta), \quad \theta \in [0, 2\pi]$$

$$\gamma'(\theta) = (-\sin \theta, 0, \cos \theta)$$

$$\|\gamma'(\theta)\| \equiv 1$$

$$\xRightarrow{\text{Pappo-G.}} \text{Area}(\Pi) = 2\pi \int_{\gamma} x \, ds = 2\pi \int_0^{2\pi} (2 + \cos \theta) \, d\theta = 2\pi (4\pi + 0) = 8\pi^2$$



Metodo 2: Param. in pelari

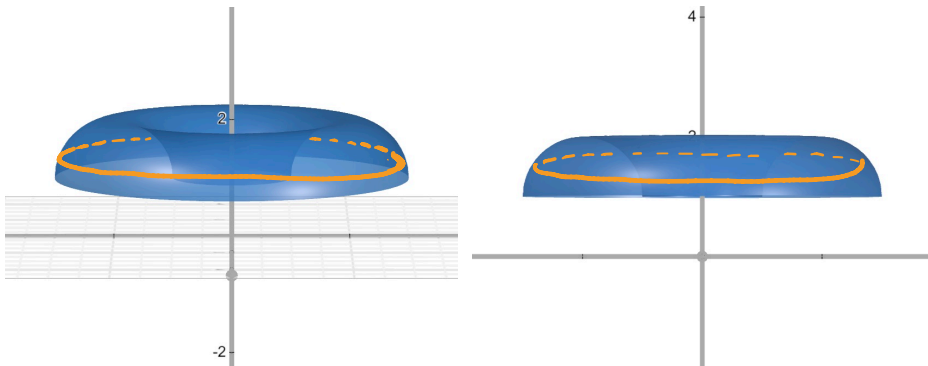
Lavoriamo su "meta toro" $M\pi := \{(x, y, z) \in \mathbb{R}^3 \mid (\sqrt{x^2 + y^2} - 2)^2 + (z-1)^2 = 1, z \geq 1\}$

$$\text{Area}(\pi) = 2 \text{Area}(M\pi).$$

$$M\pi = \Psi(D) \text{ dove } D = [1, 3] \times [0, 2\pi] \text{ e } \Psi: D \longrightarrow \mathbb{R}^3$$

$$(p, \theta) \longmapsto (p \cos \theta, p \sin \theta, 1 + \sqrt{1 - (p-2)^2})$$

$$\begin{cases} (p-2)^2 + (z-1)^2 = 1 \\ z \geq 1. \end{cases}$$



Stiamo parametrizzando le circonf. latitudinali

(Check!)

$$\text{Area}(M\pi) = \int_{M\pi} d\sigma; \quad d\sigma = \|\Psi_p \times \Psi_\theta\| dp d\theta = \underset{\substack{\uparrow \\ \langle \Psi_p, \Psi_\theta \rangle = 0 \text{ (check!)}}}{\|\Psi_p\| \|\Psi_\theta\|} dp d\theta = \frac{p}{\sqrt{1 - (p-2)^2}} dp d\theta$$

$$\text{Area}(\pi) = 2 \int_0^{2\pi} d\theta \cdot \int_1^3 \frac{p}{\sqrt{1 - (p-2)^2}} dp$$

$$= 4\pi \left[\arcsin(p-2) \right]_1^3$$

$$= 4\pi (\arcsin(-1) - \arcsin(1))$$

$$= 4\pi \left(\frac{3}{2}\pi - \frac{\pi}{2} \right) = 8\pi^2.$$

Metodo 3: Param. Contorno (grafico di funzioni).

Considerando ancora $M\pi$, ho anche

$$M(\pi) = \varphi(E) \quad \text{dove } E = \{1 \leq \sqrt{x^2+y^2} \leq 3\} \quad \text{e } \varphi: E \longrightarrow \mathbb{R}^3$$
$$(x, y, z) \longmapsto (x, y, \underbrace{1 + \sqrt{1 - (\sqrt{x^2+y^2} - 2)^2}}_{\substack{\text{ii} \\ f(x,y)}})$$

$$\text{Area}(M\pi) = \int_{M\pi} d\sigma$$

$$d\sigma = \|\varphi_x \times \varphi_y\| dx dy = \sqrt{1 + \underbrace{f_x^2 + f_y^2}_{\substack{\uparrow \\ \text{grafico di } f_z}}} dx dy = (*)$$

$$f_x^2 = \frac{x^2 (\sqrt{x^2+y^2} - 2)^2}{(x^2+y^2) (1 - (\sqrt{x^2+y^2} - 2)^2)}$$

$$f_y^2 = \frac{y^2 (\sqrt{x^2+y^2} - 2)^2}{(x^2+y^2) (1 - (\sqrt{x^2+y^2} - 2)^2)}$$

$$(*) = \frac{1}{\sqrt{1 - (\sqrt{x^2+y^2} - 2)^2}} dx dy$$

$$\text{Dunque } \text{Area}(\pi) = 2 \text{Area}(M\pi) = 2 \iint_E \frac{dx dy}{\sqrt{1 - (\sqrt{x^2+y^2} - 2)^2}} \stackrel{\text{polari}}{=} 2 \int_0^{2\pi} d\theta \cdot \int_1^3 \frac{\rho}{\sqrt{1 - (\rho - 2)^2}} d\rho \stackrel{\text{Fatto in "Metodo 2"}}{=} 8\pi^2$$

ESERCIZIO 5

Calcolare la massa superficiale di $\Sigma = \{(x, y, z) \in \mathbb{R}^3 \mid z^2 + x^2 e^{2y} - e^{-2y} = 0, 0 \leq y \leq 10\}$, avente densità di massa

$$\mu(x, y, z) = \sqrt{x^2 e^{2y} + z^2 e^{-2y} + e^{-6y} (1 + x^2 e^{4y})^2}$$

Soluzione

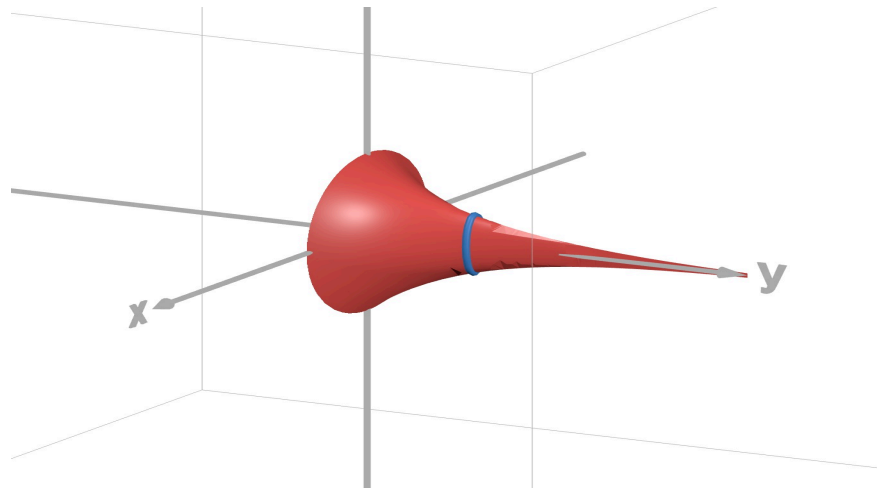
Osserviamo che $\Sigma \cap \{y = k\}$ (con $k \in (0, 10)$) è un'ellisse sul piano $y = k$, di eq

$$\frac{x^2}{e^{-4k}} + \frac{z^2}{e^{-2k}} = 1$$

Una parametrizzazione di Σ è data dunque da

$$\Psi: [0, 10] \times [0, 2\pi] \longrightarrow \mathbb{R}^3$$

$$(t, \theta) \longmapsto (e^{-2t} \cos \theta, y, e^{-t} \sin \theta)$$



Vogliamo calcolare $I := \int_{\Sigma} \mu d\sigma$, $d\sigma = \|\Psi_t \wedge \Psi_{\theta}\| dt d\theta$

$$\Psi_t \wedge \Psi_\theta = \begin{vmatrix} \dot{i} & \dot{j} & \dot{k} \\ -2e^{-2t} & 1 & -e^{-t} \sin \theta \\ -e^{-2t} \sin \theta & 0 & e^{-t} \cos \theta \end{vmatrix} = (e^{-t} \cos \theta, e^{-3t} (1 + \cos^2 \theta), e^{-2t} \sin \theta)$$

Da cui $\|\Psi_t \wedge \Psi_\theta\| = \sqrt{e^{-2t} \cos^2 \theta + e^{-6t} (1 + \cos^2 \theta)^2 + e^{-4t} \sin^2 \theta}$

$$\Rightarrow I = \iint_{[0,10] \times [0,2\pi]} \mu(x(t,\theta), y(t,\theta), z(t,\theta)) \cdot \|\Psi_t \wedge \Psi_\theta\| dt d\theta$$

$$= \int_0^{2\pi} \left(\int_0^{10} (e^{-2t} \cos^2 \theta + e^{-4t} \sin^2 \theta + e^{-6t} (1 + \cos^2 \theta)^2) dt d\theta \right)$$

$$= \underbrace{\int_0^{2\pi} \cos^2 \theta d\theta}_{\pi} \cdot \underbrace{\int_0^{10} e^{-2t} dt}_{\frac{1}{2}[1 - e^{-20}]} + \underbrace{\int_0^{2\pi} \sin^2 \theta d\theta}_{\pi} \cdot \underbrace{\int_0^{10} e^{-4t} dt}_{\frac{1}{4}[1 - e^{-40}]} + \underbrace{\int_0^{2\pi} (1 + \cos^2 \theta)^2 d\theta}_{\text{(*)}} \cdot \underbrace{\int_0^{10} e^{-6t} dt}_{\frac{1}{6}[1 - e^{-60}]}$$

Resta da calcolare

$$\begin{aligned} \int_0^{2\pi} (1 + \cos^2 \theta)^2 d\theta &= 2\pi + 2 \int_0^{2\pi} \cos^2 \theta d\theta + \int_0^{2\pi} \cos^4 \theta d\theta \\ &= 2\pi + 2\pi + \int_0^{2\pi} (1 - \sin^2 \theta) \cos^2 \theta d\theta \\ &= 4\pi + \int_0^{2\pi} \cos^2 \theta d\theta - \int_0^{2\pi} \sin^2 \theta \cos^2 \theta d\theta \\ &= 4\pi + \pi - \int_0^{2\pi} \frac{\sin^2(2\theta)}{4} d\theta \end{aligned}$$

$$\begin{aligned}
 \left. \begin{array}{l} u=2\theta \\ du=2d\theta \end{array} \right| &= 5\pi - \frac{1}{8} \int_0^{4\pi} \sin^2 u \, du \\
 &= 5\pi - \frac{2}{8} \int_0^{2\pi} \sin^2 u \, du \\
 &= 5\pi - \frac{\pi}{4} = \frac{19}{4} \pi
 \end{aligned}$$

$$\text{Finally: } I = \pi/2(1-e^{-20}) + \pi/4(1-e^{-40}) + 19\pi/24(1-e^{-60})$$